



6/17/2022

~~Explore Scientific PMC-Eight™ Firmware Update~~Explore Scientific PMC- Eight™ Software Update, March 15, 2024

~~Document Release Version 1.2
Firmware 20A01.2 Description~~

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NOTE: This document will serve as the primary source of information for the details involved with this update until the new “User Reference & Programmer’s Manual” is ready for release. This document was prepared by Jerry Hubbell and Wes McDonald for release. Please direct any comments or questions to Jerry Hubbell or Wes McDonald via the PMC-Eight Groups.io Main Forum located at MAIN@ESPMC-Eight.groups.io

PMC-Eight™ Software Release 2024 MARCH 15

PMC-Eight™ Software Release of date 2024 March 15 is available on the Explore Scientific PMC-Eight™ Software and Downloads page. The file name is **Explore Scientific Software Installation.exe**

This new install includes several files and utilities to make the PMC-Eight system better. The new download file is an executable installer. The installer will:

1. Install the newest ASCOM driver for the PMC-Eight.
2. Create a directory c:\ES PMC8 Utilities. The directory will contain the Universal Software Configuration Tool (UFCT) and a program to configure the PMC8 Wi-Fi for automatic connection to the home Wi-Fi network, making the PMC8 accessible through your home router.

A. Universal Firmware Configuration Tool (UFCT) 1.3 Changes

In addition to some nice to haves in a program like this, added the ability to program the PMC8. This will make using the old Configuration Manager unnecessary. For this capability to work, you must also have the propellent.exe and propellent.dll files in the same directory that the UFCT is in. The installer for the new driver will put those files in the directory for you.

Added is a capability to run the mount using LRDU buttons, Programming the PMC8 firmware, a means for reading documentation files, tool tips, and a refreshed look to the interface.

B. PMC8 Firmware 20A01.4.5.binary

Update to PMC8 firmware. Corrects a reported issue for indi/ekos combination brought about by erroneous transmission over the TCP or UDP link of modem configuration commands. This was reported to cause command parsing errors.

The release also corrects issues related to the Titan mount. The firmware works with the current (i.e., before this release) ASCOM driver.

This version 20A01.4.5 provides Bluetooth (BT) connectivity between the iexos100 with Bluetooth capability and a Windows PC. By using the BT, you may bypass both the Wi-Fi and serial and go directly to the PMC8 via BT. This is made possible because the PC can alias its BT connection to a COM port, which is used by the ASCOM driver. This capability works with ASCOM only, as Explorestars requires UDP Wi-Fi connectivity (ie not the COM port).

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To use this BT capability, your iexos100 must have an ESP32 Wi-Fi module that has been programmed with the BT stack. You may find out if your unit has this by running the UFCT and observing the splash screen after pressing the UFCT "Reboot and show splash" button. The module and its BT capability will be reported as NO Bluetooth or Bluetooth ON. If you have an ESP32 module but BT is not enabled, it is possible to retro fit the BT capability, although it involves opening your iexos to expose the board, installing some jumper wires, downloading and running some software to program the module. There is a procedure written for this and if you want to try it you can contact ES about obtaining it. Those who have purchased their mounts recently probably have the BT ESP32 programming already.

Older iexos100 units have a different Wi-Fi module (8266) which is not BT capable.

To use the BT capability on the PC, follow these steps:

1. After loading it into your iexos, assuming you have an iexos with BT enabled (see if the UFCT splash screen declares BT) go ahead and boot it.
2. Open device manager on the PC and find the COM port for the Standard serial over Bluetooth.
3. Open the BT settings on the PC and make sure the PMC8 is paired with the PC
4. Now open the BT setup on the PC. Find More BT options. Click on the COM tab. You will see two COM ports, one with the SPP SERVER and one for input. You will use the SPP Server comm port to connect to. Select OUTGOING as the method.

It is suggested you use the device hub. Things are a bit more responsive, but POTH also works.

5. Open the ASCOM hub settings. In the Device Hub this is done via the tools menu, and you keep selecting the ES PMC8 items until you eventually get to set the properties. It is the properties setting that finally opens the driver setup window where the COM port and other driver settings are located. Select serial but put in the COM port you found for the BT connection with the SPP SERVER. You will have to OK your way back to the main hub screen.

6. At that point you click connect. Now the first time you do this you will get a pop-up dialog box from windows asking you to connect or not. Do it.

That's it. Hub will connect and you are good to go. Once you have made these BT settings in your PC, they are remembered, and in the future all you will need to do is Connect in the hub, and the PC will, after checking in with you, hook up.

C. Configure PMC8 for Home Network:

This program allows you to configure your PMC-Eight mount to connect to your home network Wi-Fi, which lets you control it from computers attached to the home Wi-Fi, rather than directly to the PMC8. This utility can handle either the exos2, G-11, or iexos100 (that is the RN131 or the ESP32 module).

Please use the UFCT to ensure the Wi-Fi mode is set the way you want it (UDP/TCP) prior to running this program. Also, if you are going to use this with Explorestars, be sure to set the Explorestars app to use the new network address shown in the new app, rather than the 192.168.47.1 which is the normal IP. The port number remains the same.

Explorestars use still requires the PMC8 modem be in UDP mode.

This app remembers the network credentials you enter, so you won't have to enter them each time. This includes the SSID, password, and the COM port number. IMPORTANT: If you change your com port arrangement after setting the credentials (meaning have less than you used to) you may experience an error upon opening the app. The remedy is to delete the iexos IP.txt file that is found in the same directory the application is in (C:\ES PMC8 UTILITIES).

The iexos PMC8 does not remember the new network settings yet, this will require a firmware change. For now you must run this app each time you reboot the PMC8, for example between sessions. Having the firmware remember the home Wi-Fi setting may be included in the firmware in the future.

The program releases the serial port after each button press, so you may leave it open on your desktop and still use the serial port with other programs, such as the UFCT or ASCOM. Note ASCOM does not release the COM port, and the hub must be shut down before you use this program again.

It is best if you set your exos2 or G-11 to Wi-Fi channel 0 (any PMC8 that has the RN131 modem should use channel 0, but any PMC8 that has the ESP module should not be set to channel 0) by using the UFCT before switching it to your home network. Remember, if you want to go back to the PMC8 network, change that Wi-Fi channel to something else. Having the Wi-Fi channel set to 0 allows the RN131 to select whatever channel the network is present on. This is helpful if you move your mount around and have multiple access points on different channels.

If you use this program to put your G-11 or Exos2 onto your home network (that is units with the RN131 modem) it will stay there until you restore it to factory configuration. It will stay there for every subsequent boot until you restore it. This is done by running the RESTORE RN131 function in the UFCT.

Instructions:

1. Open the application. Select the correct COM port for the PMC8 serial. Set the radio button for the modem type you have in your PMC8
2. First time you run it you will have to enter your network SSID and password in the appropriate entry windows
3. You can use the GET PMC8 IP ADDRESS anytime to see what the PMC8 IP address is
4. After setting the home SSID and password, you can press the WRITE to PMC8 button to set the PMC8 up for your home network.
5. After writing the values the windows will display the PMC8 IP address assigned by your home router to the PMC8. This is the IP address you will use in your application. It will need to be put in the ASCOM driver setup for the IP address.

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In Summary, remember: The iexos will not remember the home Wi-Fi setup between boots of the PMC-Eight. The RN-131 based units such as the G-11 and EXOS2 remember the home network setting and thus to restore the unit to its native mode you must execute the RESET RN131 function in the UFCT.

D. ASCOM Driver 2024 01 16 date code on setup page

The new driver incorporates several changes.

1. It now supports variable rates. This allows you to change slew rates to arbitrary levels in programs like NINA, SkyTrack.

2. It has various bug fixes and internal improvements. Chiefly, it includes the fix for the southern hemisphere "always slewing" bug reported some time ago and corrected in the October 2022 Beta found in the files section of the subgroup.

3. It is correct for the Titan mount.

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PMC-Eight™ Firmware Release Version 20A01.2 2022 JUNE 17}

PMC-Eight™ Firmware Release date 2022 June 17 (version 20A01.2) is available on the Explore Scientific PMC-Eight™ Software and Downloads page.

Improvements to Firmware 20A01.2

1. The POTH stop button will now stop a fixed rate jog
2. Corrected failure of the mount to stop when a fixed AZ-EL move is made
3. Corrected certain conditions sometimes causing mount failure to return to sidereal rate after a slew to target
4. Corrected error that prevented a change to WiFi channel from taking place in certain conditions
5. Changed the process for responding to a Set New Configuration from the UFCT so that a reboot of the PMC-Eight is not needed, except when a user has changed the mount type.

NOTE: User must reboot and show splash manually in that case.

6. Improved accuracy of PARK in returning the mount to the HOME position
7. Internal improvements to make GoTo accuracy better
8. Added more status reporting while booting, most descriptive on a terminal program
9. Internal changes to streamline and shrink the code to enable future capability
10. PARK command now delays movement of the DEC axis for 3 seconds after RA begins moving to facilitate clearing tripod legs with cameras and eyepieces for longer scopes mounted on the EXOS2.
11. Added provisions to reduce the loss of communications connectivity to the RN131 WiFi module.
12. Addressed issue with certain tablets running ExploreStars that caused their display updates to significantly lag the mount motion.
13. Improved the accuracy of the GoTo moves with ExploreStars Point command slews.

14. Auto detects upgrade from older non-universal firmware to prevent accidental WiFi channel 0 selection after upgrade

Narrative Summary:

This version makes many improvements in the overall operation of the mount, especially regarding accuracy. Multiple internal changes were made to reduce the size of the code build to facilitate future capabilities and improve supportability. Various changes were made to improve the user experience when employing ExploreStars and/or ASCOM.

Associated with this firmware release are changes to the Universal Firmware Configuration Tool and the PMC-Eight™ ASCOM driver. The UFCT now will warn the user if the WiFi channel is set to 0. This channel is appropriate for the RN131 ONLY and is not appropriate unless the user has manually programmed the RN131 to connect to their home network. The Driver has improvements that will make GoTo accuracy better than ever. Our experience is the pointing (in a mathematically perfect coordinate system, absent the various mount setup and mechanical errors that are common and difficult to get to zero) is typically <15 arc-seconds for large slews and considerably less for short slews. These changes will make those plate solve slews to target typically to a few arc-seconds after one solve. Discrepancies reported for guiding performance that were driver related have also been corrected.

The firmware, Driver, and UFCT now support the Titan mount. Titan owners should update their firmware, Driver, and UFCT to this release version. These also support a special mount which users will note is called Scotty. This is a special configuration and is not for general use, DO NOT SELECT AND USE THIS MOUNT TYPE. Users are encouraged to install this updated version, however as always if their current system software meets their needs, they may choose to continue using their existing versions.

The new UFCT and new Driver are compatible with the previous firmware release 20A01.1 but the stop button fix will not work. The previous version of the driver is completely compatible with the new firmware version 20A01.2. It is recommended that the new firmware be installed coincident with the new driver.

20A01.2 Firmware Code Fixed Notes:

09OCT2021

1. Corrected subtle bug detecting ESP32 (reversed key letters in search)
2. Fixed bug in wait_ra_point that prevented the mount from returning to sidereal when a re-directed slew was attempted
3. Fixed bug in wait_ra_point caused by 6. that made AZ-EL move not stop RA
4. Changed MINSPD to allow setting rate to 0 with ESSr
5. Added ES2 calls for mount movement from secondary ES command language user
6. Removed wifi capability for Diagnostic, now only serial

10JAN2022

1. Replaced Old Adjust_Mx_PARAMS routines with single method using indirect addresses
2. Replaced old SET_Mx_MICRO routines with single method using indirect addressing
3. More changes to ESSxPreamble to fix interrupted gotos - not right yet
4. Added more visual cues for boot process, code shrink
5. Added ESV! command to request full up mount state vector in one request
6. Fixed issue that prevented unit from working with 8266 TCP

22JAN2022

1. Combined GOTO_RA_POINT and GOTO_DEC_POINT into a single method GOTOMotor_POINT for code efficiency/maintainability
2. . Modified PAUSE routine for code efficiency
3. Altered timing of wifi modem module to address busy p... messages with 8266
4. Corrected INITESP to prevent sending BLE setups for the case of older ESP32 firmware

28JAN2022

1. Added lock outs for secondary ascom user. Secondary ES commands are those for ES2

05FEB2022

1. Changed Command \$16 to be for a broken ESSi command. \$06 is now used for echo of ESGi post ESSi. No auto reboot after ESSi

12FEB2022

1. ESPt3 change broke POTH rate move stop button because rate change did not use a Pt command. Now call rampdown with Pt3 statefully

15FEB2022

1. Fixed bug in DEC_RAMP_DOWN with an indent in repeat loop seemed to work anyway

16FEB2022

1. Corrected failure to reload wifi channel at command \$06 after ESSi command

23FEB2022

1. Changed \$06 back to only be used for echo of ESGi. Put other functions into \$16, made \$15 error return from Serial mirror for ESSi
2. Added TCP/UDP and motor current immediate change to \$16

24FEB2022

1. Blocked sending of BLEADVER message when in ST4 diagnostic print mode
2. Made ST4 enable/disable flag be immediate changed in \$16
3. Made P mode in ExploreStars resume sideral rate in RA between point commands
4. Shortened time delays during RESET_UDP routine, made it go to WIFI port only

02MAR2022

1. Used case 10 in cog 4 instead of IF statement for ST4 cases
2. Corrected ASCOM PARK not parking on first attempt, this by increasing ramp intervals to 400 from 200

05APR2022

1. Altered Wait_RA_Point to provide a finishing move if target is 0,0 OR ESPt_STOP. Ensures exact park location when park <> home

19APR2022

1. Restored STOP after each P mode move to make az-el coordinate move stick without dither (may want to not stop for smooth P mode tracking, but rates are constantly changing)

05MAY2022

1. Corrected up ramp start speed to be zero, corrected old bug where 0 was thought to be but wasn't

27MAY2022

1. Added support for Titan mount as P9 = 13
2. Added a Restore Comms call every 20 seconds for RN131

31MAY2022

1. Reduced Out inquires to every other one to resolve tablets getting behind on display updates
2. Made JOC RA point commands continue at sidereal after point command

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3. Added 2* sideral motor counts to JOC RA point commands, this corrects for an error caused by 2 second no ramp move. Also compensated a bit more for comms delays, significant accuracy improvement for GoTo

7JUN2022

1. Used S parameters to reject WiFi channel 0 on first factory boot or upgrade from non-universal version

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PMC-Eight™ Firmware/Driver/Tool QuickStart Installation Guide – See guide starting on Page 13 below.

For installation instructions please refer to the instructions below that were written for version 1.1. Note the file names etc. needed for this version are altered to reflect the revision codes. Aside from that the instructions remain the same, as does the basic operation of the firmware.

Users may employ ASCOM platform 6.4 or newer. Both POTH and the ASCOM Hub have been used in testing and either may be employed.

NOTE: THE ASCOM POTH is available on the platform installer, be sure to select it when running the platform installation program

SOFTWARE DOWNLOAD (UPDATED TO VERSION 20A01.1)

1. Download the zip file containing all the files needed from the Files section of the PMC-Eight MAIN Groups.io Forum. Look for “PMC-Eight 20A01 Firmware Installation Kit 2022-06-17”
[PMC-Eight 20A01 Firmware Installation Kit 2021-08-08](#)
2. Unzip the file into a folder of your choice, make sure all the following files are available in your chosen directory:
 - a. PMC-Eight™ Universal Firmware Configuration Tool (UFCT) Executable file (2021-06-17).
 - b. PMC-Eight™ Configuration Manager (CM) 1.03 Installation Executable file (2021-05-13).
 - c. PMC-Eight™ Version 20A01 Universal Firmware EEPROM file (2022-06-17).
 - d. PMC-Eight™ ASCOM Driver Version (2022-06-17)
 - e. PMC-Eight™ Firmware Release Notes PDF File.
3. Unzip and run the installation program for the CM.
4. Copy the PMC-Eight™ UFCT Executable to a location of your choice and create a shortcut to start it from the Windows Desktop.
5. Refer to Page 13 to start installation of new firmware and Driver.

PMC-Eight™ Firmware Version 2.0 (20A01.1 Release Version 2021 AUG 08)

PMC-Eight™ Firmware Release dated 2021 August 08 (version 20A01.1) is available on the Explore Scientific PMC-Eight Software and Downloads page.

This version includes the following major updates and changes:

- Simultaneous Communications Enabled Using Both Serial and WiFi Channels (ASCOM and ExploreStars at the same time)
- Added Restore RN131 Configuration Button to the Universal Firmware Configuration Tool (UFCT) to make it easy to restore a broken manual RN131 configuration installed via the procedure.
- Miscellaneous fixes to the UFCT and the ASCOM Driver identified over the past 4 months.

Narrative Summary:

This version fixes minor issues with intermittent hangs and timing issues with the real-time operation of the motor driver processes. Fixed intermittent process hangs when using the manual pushbuttons on ASCOM clients, i.e., POTH, and N.I.N.A. Added a function to smoothly ramp down the mount when stopping. Improved the mount pointing precision when going to objects and to the HOME (default PARK) position. Completed and fully tested the code used to provide simultaneous communications when connected to the SERIAL Port and WiFi Communications channels. Updated and improved the operation of the ASCOM Driver to make it more reliable.

Firmware Code Notes:

- | | | |
|------|-----------|---|
| 1.01 | 13APR2021 | 1. Changed watchdog timeout in COG 2 equal to COG1 |
| 1.01 | 05May2021 | 2. Fixed bug in cog1 final waitcnt that sometimes overran |
| 1.01 | 05May2021 | 3. Fixed lack of commit following set in five places |
| 1.01 | 11May2021 | 4. Fixed issue with RA Motor driver curcnt wrapping stopping motor. |
| 1.02 | 16May2021 | 5. Fixed problem with duration value miscalculation causing manual pushbutton ESSr commands hanging up with another wrap around issue. Removed all dead code and cleaned up formatting in places. Removed unused logical tests. |
| 1.03 | 20MAY2021 | 1. Implemented ESpt3! command to stop mount current goto slew sequence |
| 1.04 | 16JUN2021 | 1. Added ESSr_LOCK and associated state machine to make LRDU work correctly in simultaneous ASCOM/ExploreStars operation |
| 1.05 | 12JUL2021 | 1. Fixed issue with waitrapoint() that caused slews to home not to be precise
2. changed to an ES OK echo for redundant changes into or out of debug mode to differentiate from modem replies
3. Made continuous track mode default on with fresh system load
4. Added passthrough command to interpreter
5. allow WiFi channel 0 for infrastructure mode on RN131. Must change to 0 via diagnostic P command, ESSw will not accept |
| 1.1 | 25JUL2021 | 1. Initialized temp_dir before M2 repeat loop (clean up code omission)
2. Candidate Release 1.1 |

PMC-Eight™ Universal Firmware Version (20A01.0 Release Version 2021 APR 12)

The new PMC-Eight™ Universal Firmware version is the result of thinking about the system for the past 3+ years of operational experience since the release of the PMC-Eight in 2017. Our team at Explore Scientific considered our own and most importantly, the experiences of the hundreds of customers using their PMC-Eight™ Control System in the field. After much discussion and reflection about the system and how it should evolve into the future, the following changes and improvements have been implemented in the new firmware version.

I want to give a special thanks to Wes McDonald for his tireless work over the past 6 months in bringing this firmware to fruition and his countless suggestions on improvements and changes to the firmware.

New and Enhanced PMC-Eight™ System Features, Behaviors, and Parameters

New Universal Firmware Version Applicable to All Currently Delivered, and Future Mounts

This version of the firmware is designated the “Universal Version” in that the compiled binary loaded into the controller’s EEPROM can be loaded on “any” PMC-Eight™ mount system and subsequently be configured for that specific mount. Having a Universal Version of the firmware means that there is only one codebase to maintain instead of having several versions to maintain for each mount/hardware system. The main difference between the previous firmware versions was based on the need to provide different motor currents to each of our different mounts using different motors and gear drives.

New “Always On” and Simultaneous WiFi/Serial Port Communications Functionality

One of the most significant updates to the PMC-Eight™ control system is the enabling of “Always on Communication Channels”. This functionality is designed to make it much easier for the customer to use the PMC-Eight™ via the serial port or via WiFi without any additional configuration changes. This has been a source of frustration for a number of customers over the last few years.

Additionally, changes to the firmware have been made to also allow “Simultaneous Channel Communications”. The legacy “ESX!” command has been deprecated and is no longer needed. To accomplish this, the concept of “Command Modes” is introduced as a consequence of the need to provide a “Command Lockout” feature to manage concurrent communications through both channels. There are 3 Command Modes: “Monitor”, “Execute”, and “Diagnostic”. Each of these modes help to define which commands are allowed for each mode. The Command Types are: Monitor Commands, Execute Commands, and Diagnostic Commands.

The basic functionality is that when a host sends an Execute Command, if the command processor does not have a current lock on the channel that the host is currently using, then a “Command Lockout” will be generated on the opposite communications channel. Once the command has finished executing, then the lock will be released. This process is mainly used to marshal the communications when both the serial port and the WiFi port are actively being used. The typical case will be when using the PMC-Eight™ ASCOM Driver on the serial port along with the ExploreStars™ application on the WiFi port. The result is that the system behaves and provides a “Bumpless Transfer” of command authority and real-time command processing between hosts. As an example of this operation, when an ASCOM device requests the mount be Slewed to a position the PMC-Eight™ will lockout and ignore all Execute Commands from the ExploreStars™ application. This is primarily when ExploreStars™ is in the POINT MODE. This will continue indefinitely until the user puts the ExploreStars™ application in TRACK MODE. At that time, the PMC-Eight™ will recognize and respond to ExploreStars™ issuing Execute Commands.

New System Hardware Features Detection on Boot

The firmware includes an automatic hardware features detection algorithm that determines two main hardware components. The type of WiFi module (ESP, or RN131) is automatically determined on boot and configured accordingly. The Type of ST4 port is also determined: the legacy analog ST4, or the current digital ST4 port. Other configuration parameters (P Values) are determined at boot time based on the mount type set in the P9 parameter (see details below).

Updated System Status LED Flash Modes

Commented [WM1]: You noted the use of a hand paddle! Oops.

Commented [WM2]: This whole section is sort of clunky and deep but it is ok for now. Most people will just ignore all this.

The Process LED provides feed back to the operator concerning how the PMC-Eight™ is operating by providing indicative flashing sequences. The Process LED Flashing Modes on boot/startup and during operations are as follows:

Fast Flash Mode – Communications Watchdog Timer Disabled – This signifies that the mount will continue to track even when communications is lost from the host computer. When enabled, will be indicated whenever the mount is not being communicated with by an application, during which time the mount will continue to track the RA axis at the correct sidereal rate.

Slow Flash Mode – “Motor Mode” – This signifies that the system will start tracking automatically on power up with no host connected to the system. When enabled, motor mode will move the RA axis at the correct sidereal rate and in the direction required by the Northern or Southern Hemisphere mount location as set by the UFCT. Note that once communications occur, the motor mode ceases until the next PMC-Eight™ boot.

Continuous on Mode – This is the normal mode when active communication is occurring between the system and the host computer. The LED will be steady on when communications are occurring at a regular interval.

Single/Double Flash Mode on Startup – This mode shows the current configuration of the WiFi protocol stack. Single flash is UDP, and a double flash is TCP. It always happens upon PMC8 bootup, after which the other flash modes apply.

Enhanced Pointing Capability and Tracking Accuracy

A new level of pointing capability has been established with this version of the firmware. The ASCOM driver “SlewToTarget” API commands, working with the new firmware motor driver timing improvements has resulted in an enhancement to the system pointing capability. This ensures that the only limits to pointing precision are based on the performance of the hardware and its manufacturing quality. The tracking rates have also been tuned to provide the best possible tracking precision when using the mount for visual observing or astrophotography.

Communications Watchdog Toggle (Enable/Disable)

The new “ESW!” toggle command allows for easy switching between protecting the mount when communications drops (Serial or WiFi) or allowing the system to continue tracking without the need to have the host computer connected when doing standalone astrophotography with a DSLR system. This command when issued through a terminal program will set the mount into continuous track mode, after which the status LED Fast Flash discussed above will apply.

Enhanced “Dongle Insert” Functionality

In the previous firmware the “Dongle” or WiFi Channel Changing Device (WCCD) is used to increment the WiFi channel while in the field.

The WCCD has added functionality in the new firmware and changes the operation of WiFi channel change.

1. When using the PMC-Eight™ in the field it is sometimes necessary to toggle the WiFi mode between TCP and UDP. In the past this required having a PC. With the new firmware, if the dongle is inserted prior to powering the system, upon power up the dongle will trigger a toggle of the WiFi protocol. This is especially useful for users who find themselves in the field trying to use the ExploreStars™ application after having inadvertently left their system configured for ASCOM use with TCP.
2. While the PMC8 is operating, the user may increment the WiFi channel by inserting and removing the dongle. Each time the dongle is inserted and removed the channel is incremented by one and may be performed as many times as needed without reboot of the PMC-Eight™. This constitutes a change from the old firmware where only one change was allowed, and a reboot was required between changes.

New Auto-Tracking on Boot

The new “ESM!” toggle command allows you to startup your mount without any device connected to the mount. Enabling this function uses the default Sidereal Tracking Rate value for the mount determined through the Mount Type selected and stored by the UFCT (stored in the P9 value). This is great for those who do not want to use GOTO and just want to turn on the mount and manually position the telescope to different objects in the sky. This is identical to the legacy operation of a mount system prior to the availability of motor driven GOTO mounts in the last century.

Commented [WM3]: Oops!!

Commented [WM4]: Fine for the programmers manual, not fine for this document

New System Bootup Hemisphere Selection

The UFCT allows the user to set the mount hemisphere for Northern or Southern. This is only important when using the stand alone Auto Tracking but should be set correctly by the user in all events. The new terminal “ESH!” toggle command provides another method to select the tracking direction when using the Auto Tracking feature “ESM!” command. You can select either Northern or Southern Hemisphere.

Enhanced Sidereal Rate Fraction Setting

The UFCT allows the user to set the PMC-Eight™ sidereal rate fractions for guiding, now with different values in each axis. New terminal commands, t “ESSfA...!” and “ESGfA!”, where A is equal to the Axis, has been enhanced to allow you to set different values for the Sidereal Rate Fraction (SRF) for the Right Ascension and Declination axes. This value can be changed “on the fly” via the ASCOM driver and will immediately be applied to not only the ASCOM API PulseGuide Method, but also on the hardware/firmware based ST4 Autoguiding port.

New Factory Initialization and Configuration Storage Functions

At the factory, on initial boot after firmware load, the EEPROM P Values are not set and all have a value of 0xFFFF (-1). The system detects this and sets default values for specific parameters including P0, and P2. After that P9 is set manually using the Diagnostic Command Mode, and on subsequent boots will set the default motor current limits and the initial motor current values. On subsequent boots of the system, the values will be bounds checked to ensure they are still within specifications prior to system startup.

Configuration Features and Values

Commented [WM5]: Not sure any of this is needed in this document.. Programmers manual yes.

Initialize P Values

Bounds Check of P Values on Boot

EEPROM Parameter Values (P Values) (Some values have dedicated ES Commands as indicated)

P0 – BAUD Rate (*Default Rate)

9600
14400
19200
38400
57600
115200*

Enumerated Value Checked

P1 – WiFi Channel Number (Deprecated Communications Channel ID)

Range 1-11

P2 – Network Protocol (“ESY!” Toggle)

0 – UDP/IP
1 – TCP/IP

P3 – NOT USED

Future

P4 – Communications Watchdog Disable (“ESW!” Toggle)

0 – Not Enabled
1 – Enabled

P5 – Boot Tracking Enable (“ESM!” Toggle)

0 – Not Enabled
1 – Enabled

P6 – Bootup Hemisphere (North/South) (“ESH!” Toggle) Used in Conjunction with P5 (“ESW!” Toggle)

0 – Southern Hemisphere
1 – Northern Hemisphere

P7 – RA Sidereal Rate Fraction for ST4 Port (“ESGf0!”, “ESSf0...!”)

Range From 1 to 100 (Percent)
Range Checked

P8 – DEC Sidereal Rate Fraction for ST4 Port (“ESGf1!”, “ESSf1...!”)

Range From 1 to 100 (Percent)
Range Checked

P9 – Mount Hardware Defined Values (Manually Set in Diagnostic Mode)

0x0 – iEXOS 100 w/ESP8266 (1A01B/C)
0x1 – iEXOS 100 w/ESP32 (1A02A)
0x2 – iEXOS 200 w/ESP32 (1A02A)

0x3 – iEXOS 300 w/ESP32 DB (2A07A)
0x4 – G11 w/RN131 DB (2A06B)
0x5 – G11 w/ESP32 DB (2A06B)
0x6 – G11 w/RN131 DB (2A07A)
0x7 – G11 w/ESP32 DB (2A07A)
0x8 – EXOS2 w/RN131 DB (2A06B)
0x9 – EXOS2 w/ESP32 DB (2A06B)
0xA – EXOS2 w/RN131 DB (2A07A)
0xB – EXOS2 w/ESP32 DB (2A07A)

PA – NOT USED

PB – ST4 Port Print Out – In Diagnostic Mode print out ST4 port current input value

PC – ST4 Port Disable – 0 is Enabled (FALSE), 1 is Disabled (TRUE)

Mount Type – P9 Value Defined

Autoguider Type (ST4) Analog/Digital

WiFi Module Type (Microchip RN-131/ESP8266/ESP32)

Value is Key for Mount Current Limit Settings

New Customer Set Motor Current Settings and Factory Motor Current Limits

To accommodate all the different mounts that the PMC-Eight™ control system can be used on, it is important to manage the motor current supplied to each separate motor drive system. Each mount has its specific motor drive system consisting of a specific stepper motor size and gear/pulley drive system. This is accomplished by creating a set of motor current limits applicable to each motor/drive configuration. Currently there are 3 defined sets of limits as shown below. The user can set two Motor Current values that are bound by the minimum and maximum “S Values”; Motor Track Current Value, and Motor Slew Current Value. These two values can be set to provide the best battery life, the highest torque, or anywhere in between the applicable limits.

Mount Specific Limits – Chosen and Used Based on P9 Value

CONSTANTS (Hard Limits) – Default Factory Values

Two Values – Minimum and Maximum

EEPROM Limit Values (S Values) Assigned to Specific Mount Class

These limits are “hardwired” in the firmware and are not changeable by the user. These are **determined** by the size of the motor in the mount, by the testing performed at the factory, and by using feedback from our customers over time. The value for Tracking and Slewing can be set to any value within the range for the specific mount.

S0 – Minimum Current 0 (G11) 600mA
S1 – Maximum Current 0 (G11) 1300mA
S2 – Minimum Current 1 (iEXOS 100) 400mA
S3 – Maximum Current 1 (iEXOS 100) 900mA
S4 – Minimum Current 2 (EXOS2) 400mA
S5 – Maximum Current 2 (EXOS2) 900mA
S6 – Minimum Current 3 (iEXOS 200) 400mA

Commented [WM6]: This is not strictly speaking correct. The S values are changeable of course and once set by a knowledgeable operator are the applicable limits independent of the firmware. Thus we can instruct someone to nudge one higher. I think. Maybe not though because setting them might be limit checked. Anyhow they are changeable somewhat by the user. But if we have any of this in this manual then we can leave it the way it is for further explanation later as needed, since we don't want them playing with these any how

S7 – Maximum Current 3 (iEXOS 200) 900mA
S8 – Minimum Current 4 (iEXOS 300) 600mA
S9 – Maximum Current 4 (iEXOS 300) 1300mA

There are 2 Motor Current Settings that are limited in value – Track Current and Slew Current. These may be set by the UFCT. New terminal commands have been added also:

“ESScAXXX!” Allows the user to set the Motor Current Values for Track (A=0) and Slew(A=1) within the defined limits. “ESGcA!” Allows the user to Get the Current Motor Current Settings for Track(A=0) and Slew(A=1).

PMC-Eight™ System Operation Use-Cases

The PMC-Eight™ firmware allows for several different system use configurations involving single or concurrent multiple host communications and/or manual operations with ASCOM and ExploreStars™. The communications system handles all host requests and responses to provide a good user experience when issuing commands from more than one host.

Different Use-Case Types:

ASCOM (Serial/WiFi) Standalone

Great for dedicated host computer use, especially when remote access is the goal

ExploreStars™ (WiFi) Standalone

Dedicated Communications Mode

Automatic Control Switching and Lockout

Great when using ExploreStars™ as the primary hand-paddle control system for manually adjusting the mount and using the ExploreStars™ GOTO and Alignment functions.

ASCOM (Serial) + ExploreStars™ (WiFi)

Concurrent Communications Mode

Automatic Control Switching and Lockout

Great for remote observatory control with a local computer system along with using ExploreStars™ as the hand-paddle system for manually adjusting the mount when at the mount.

DEFINITIONS

Command Lockout – This is the action of the PMC-Eight™ system responding to a request from a given host. If the command is an “Execute Command” then the system will lockout the other channel and ignore any “Execute Commands” coming from the other channel while the current command is being processed.

Command Mode – This is one of three types of commands that the PMC-Eight™ is designed to respond to in the context of concurrent connection operations. The three types of commands are Execute, Monitor, and Diagnostic. Execute Commands are priority commands that lockout the other channel from that issuing the command. The other channel, once locked out, can only expect responses from Monitor Commands. The Diagnostic Mode is a “test” mode and will disable any concurrent communications and

Commented [WM7]: If you blow out the stuff about the S parameters, retain this little bit.

Commented [WM8]: Oops!!!!

restrict commands to only Diagnostic Commands coming from the host issuing the request to go into Diagnostic Mode via the “%%” command.

“Always On” Communication Channels – There is no longer any need to explicitly switch between the “serial” or “wireless” communications channels, they are both available to make a connection at any time.

Concurrent Connection – When two hosts (ASCOM/INDI, or ExploreStars™) are connected to the PMC-Eight™ system using both the Serial and the WiFi communications channels. This connection allows simultaneous control of the mount. There are specific limiting conditions for operation (LCOs) that allow this to occur but bounds the user in certain ways to ensure reliability of the operation. Generally, the serial channel controller has priority over the wireless channel controller.

P Values (Parameters) – The different defined configuration parameter values stored in the PMC-Eight™ system EEPROM memory. These values are set by the system during the boot process under certain circumstances or can be set manually using the “Diagnostic Command Mode”.

S Values (Setpoints) – These values are used to store the different “Mount Current Limits” defined at the factory, and prepared for setting on the system on initial boot. These values are automatically set by the system during the boot process unless changed by the user using the “ESSc0...!”, and “ESSc1...!” commands.

Sidereal Rate Fraction (SRF) – The SRF is used by the Autoguider system to help calculate the correction value to return the mount to the target coordinates. There are 3 parameters used to correct for the PE motion of the mount “off-target”: SRF, Pulse Time, and Direction. The equation to correct for a given measured offset in arc-sec is the SRF times the Pulse Time. i.e., (SRF x 15.00) arc-sec/second x Pulse Time seconds. The result is the amount of movement in arc-sec to put the mount back on target.

WiFi Channel Selection – The ES command “ESGwXX!”, and “ESSwXX!” can be used to directly set the WiFi channel and store it in EEPROM. The valid values for channel are 1 to 11. The PMC-Eight™ “Channel Changing Device” or dongle as it is sometimes referred to, can still be used to increment the channel when in the field without a host computer connection.

PMC-Eight™ Firmware/Driver/Tool QuickStart Installation Guide

Here are the basic steps to install the new firmware, ASCOM driver, and Configuration Tool:

NOTE: Although you can continue to use ASCOM Platform 6.4 SPX, we suggest that you download and install the current release of the ASCOM Standards [Platform](#) (Version 6.5). It is strongly suggested that when you install the ASCOM platform, that you select and install the POTH Hub for use with the PMC-Eight™ ASCOM Driver versus using the Device Hub which has demonstrated a few problems during testing.

Commented [WM9]: Is platform 6.5 required? Also are going to mention they should continue to use POTH of now?

SOFTWARE DOWNLOAD (UPDATED TO VERSION 20A01.1)

1. Download the zip file containing all the files needed from the Files section of the PMC-Eight MAIN Groups.io Forum. Look for “PMC-Eight 20A01 Firmware Installation Kit 2021-08-08”
[PMC-Eight 20A01 Firmware Installation Kit 2021-08-08](#)

Explore Scientific PMC-Eight™ [Firmware-Software](#) (Version 20A01.4.52) Release Notes
Document RELEASE Ver. [1.4.5 2024 3 15 17-2022-06-17](#)

2. Unzip the file into a folder of your choice, make sure all the following files are available in your chosen directory:
 - a. PMC-Eight™ Universal Firmware Configuration Tool (UFCT) Executable file (2021-08-08).
 - b. PMC-Eight™ Configuration Manager (CM) 1.03 Installation Executable file (2021-05-13).
 - c. PMC-Eight™ Version 20A01 Universal Firmware EEPROM file (2021-08-08).
 - d. PMC-Eight™ ASCOM Driver Version (2021-08-08)
 - e. PMC-Eight™ Firmware Release Notes PDF File.
3. Unzip and run the installation program for the CM.
4. Copy the PMC-Eight™ UFCT Executable to a location of your choice and create a shortcut to start it from the Windows Desktop.

FIRMWARE INSTALLATION

1. Using the CM Advanced Tab, Select the new universal firmware file 20A01.1.eeprom to install on your PMC-Eight.

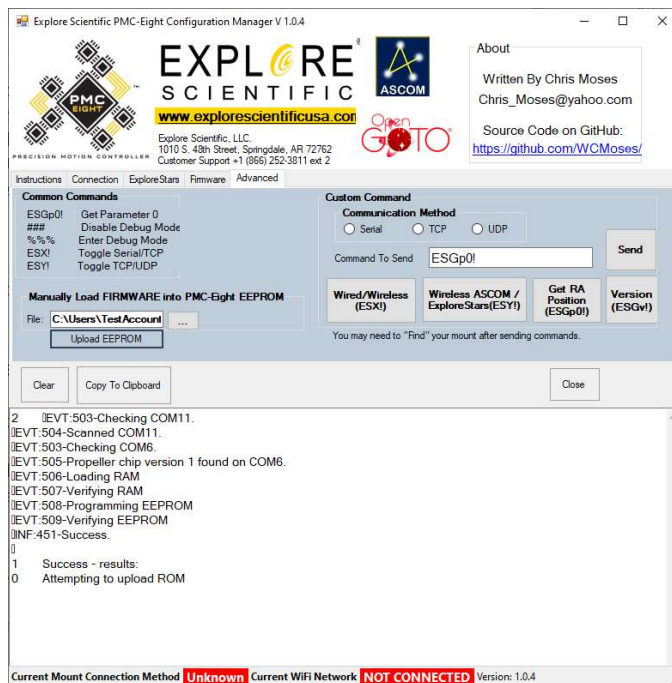


Figure 1. PMC-Eight™ Configuration Manager Firmware Upload

2. Push the Upload EEPROM button to manually install the new firmware.
3. Verify the proper successful completion of the installation via the log messages as shown in figure above.

PMC-Eight™ ASCOM DRIVER INSTALLATION AND CONFIGURATION

NOTE: While the new PMC-Eight™ ASCOM Driver version 2021-08-08 is substantially the same as the previous versions released in 2019 and 2020, it contains multiple improvements and changes necessary for compatibility with the new Universal Firmware Version 20A01.1 it is strongly recommended that you install the new driver. Similarly, the new driver should not be used with the older versions of the firmware, as some results will be unpredictable.

When the new driver is installed, the old one should be removed completely from your Windows system. This should be automatic for most Windows systems, but there have been instances where removal of the previous version has not occurred correctly. The best way to ensure proper removal of the previous driver is to manually delete the driver DLL file, and to clear out the ASCOM driver profile using the ASCOM Platform Profile Explorer as follows.

1. Before you remove the existing driver, you may wish to make note of your existing driver configuration settings (for example your site location) as these will be replaced with defaults when the new driver is installed.
2. Remove the existing driver profile by starting the ASCOM Profile Explorer and removing the highlighted Telescope Driver shown below in Figure 2 – ASCOM.ES_PMC8.Telescope. Just right click on it and Select “Del Key”

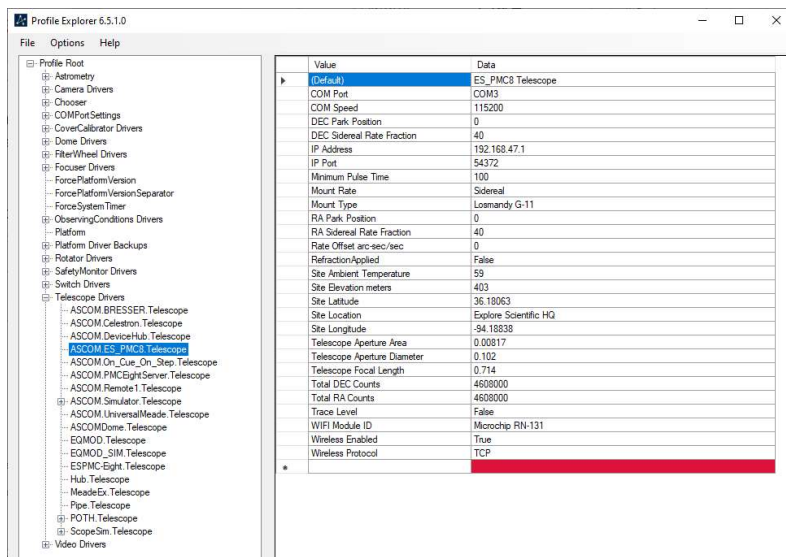


Figure 2. ASCOM Profile Explorer.

3. Navigate to the directory: C:\Program Files (x86)\Common Files\ASCOM\Telescope and delete or rename the file: ASCOM.ES_PMC8.Telescope.dll from the directory.
4. Run the new PMC-Eight™ ASCOM Driver Installation Program to install the new driver.
5. The new driver will replace your previous driver settings with default values. These must be reset by you through the normal ASCOM settings chooser before you connect the driver to the PMC-Eight™.
6. After installing the new ASCOM Driver, you should run the Profile Explorer again and compare the settings to those shown in Figure 2. Note that there are now two Sidereal Rate Fraction Values, one each for RA and DEC.
- 7.

FIRMWARE CONFIGURATION

1. The new firmware must be configured prior to operation of your equipment. This is done by using the new PMC-Eight UFCT. Use this program to select the features and parameter values desired and to set the configuration to match your mount's hardware configuration. See the next section for a description of the UFCT.
2. Connect the PMC8 controller to the host PC and boot the UFCT. Follow the steps shown in the log window (A-D). When the SET New Configuration button is pressed there may be some delay where the log message of Busy Please wait appears multiple times as the PMC8 reboots. PMC8 booting requires a bit longer than previously. The Busy message will persist for several message repeats. If the Busy message repeats excessively, reboot the PMC8 manually.

New PMC-Eight Universal Firmware Configuration Tool 1.1 (UFCT)

A new software tool (Microsoft Windows Only) is provided to configure existing customer PMC-Eight™ Mount Control Systems for use with the new Universal Firmware Version 20A01. This interface provides a way to set all the features included in this new firmware and provides a terminal interface to manually send commands to the system and save the terminal log as a file (Figure 3.)

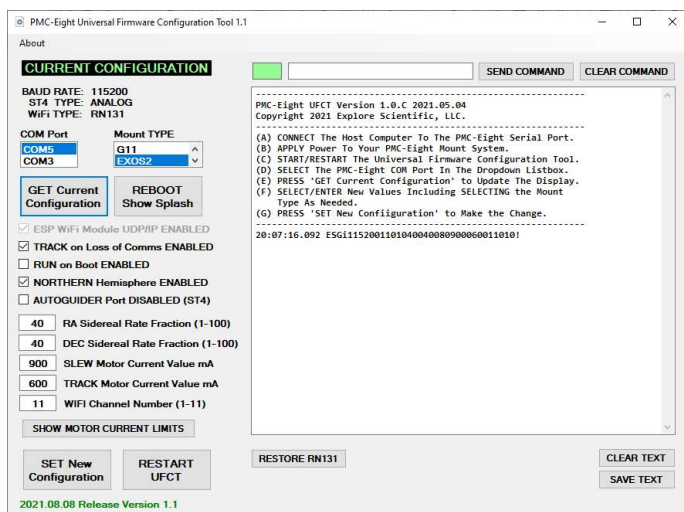


Figure 3. PMC-Eight Universal Firmware Configuration Tool 1.1

NOTE: The UFCT is only applicable to the Universal Version of the Firmware (20A01) and later and is only used to configure the PMC-Eight™ hardware parameters as stored in the PMC-Eight™ Controller.

NOTE: When you press GET CONFIGURATION the UFCT presents the settings as they are internal to the PMC-Eight™. Most of these settings did not exist on the old firmware and thus upon first boot of the new firmware, these settings are placed into usable default values. The PMC-Eight™ controller has no way to know if these settings are what you want, and thus they should all be set by you.

NO CHANGES YOU MAKE ON THE UFCT INTERFACE ARE SAVED TO THE PMC-Eight™ UNTIL YOU PRESS the SET New CONFIGURATION button!! This includes, and is especially important for, the mount type setting.

IMPORTANT NOTICE ABOUT THE WIFI CHANNEL: When the Universal Firmware is first loaded into a PMC-Eight unit previously programmed with the older legacy firmware, the WiFi channel that is initially set will be 0. This channel is not going to connect to the PC WiFi. It will need to be set using the new UFCT to a channel between 1 and 11. Using one of channels 1, 6, or 11 is recommended. Users who want to experiment with setting their EXOS2 or G11 PMC-Eight units to connect to their home networks will want to use the WiFi channel 0 setting, all others must not. Be aware that setting the WiFi channel to 0 is not in itself sufficient to get the PMC-Eight connected to the home network. The iexos100 cannot make use of WiFi channel 0 at all.